# L2 Norm Predictor: Points an Examiner Could Attack

**Thesis:** A Unified Benchmark of Grokking Predictors in Neural Networks · Predictor 1 (L2 Norm) · Prepared 5 November 2026

## Current state of L2

L2 Norm is Predictor 1. It runs on the final Nanda-Unified setup (`nanda_unified.yaml`, $p = 113$, 1-layer 4-head transformer, small init, AdamW with weight decay $\lambda = 1.0$ and betas $(0.9, 0.98)$, 5 seeds, 40,000 epochs, results committed on 25 September 2026). It tracks the total weight norm $\|w\|_2 = \sqrt{\sum w^2}$ every epoch and uses two detection rules on log-epoch moving averages of it: the fast/slow crossover (windows 50/200) and the MA-of-MA zero crossing (window 20, first 100 epochs skipped). Both settings are fixed in the config. L2 was closed as negative on 26 September 2026 because both rules fire at almost the same early epoch in every seed. The "halfway down from peak" reading is recorded as a hindsight check rather than a benchmark rule, and the frozen-protocol verdict hasn't been saved to a results file yet.

This document lists weak points in the final L2 Norm implementation and experiment that a thesis examiner could raise. Each point says what was done and why it could be challenged. Solutions are deliberately not included. Points are ordered from most to least serious.

## Big worries

**1. L2's official result was never scored under the frozen protocol, and no rule is named as official** L2 was closed on 26 September using a working two-part test (fires before grok in all seeds, and $\rho$ across seeds). The frozen three-part rule (lead_frac $\geq 0.8$, $\rho \geq 0.9$, $0.05 \leq$ mean relative lead $\leq 0.9$) was only written on 29 September. On 2 October, L2 was deliberately left out of the scored list because its module never named one rule as the official one, and saving its verdict is still listed as next work. Even so, the 27 October and 4 November progress reports list L2 alongside the predictors closed "under the frozen protocol", which reads as if all eight were scored the same way. When the frozen rule is applied to the saved data, both implemented rules fail: the crossover gets lead_frac $= 1.0$, $\rho = 0.70$ and mean relative lead $0.989$, and the MA-of-MA crossing gets $1.0$, $\rho = 0.00$ and $0.835$. An examiner can still ask which rule is L2's result and where that verdict is recorded.

**2. The record doesn't support "the test was fixed before Predictor 1"** The viva notes say the test was pre-registered before Predictor 1 and applied unchanged. For L2, the record shows three different tests. In August it was three criteria, one of them a noise-floor check. On 26 September it was a two-part working test, and on 29 September came the frozen three-part rule. Its $\rho$ cutoff was calibrated using this run's own grok epochs, after L2's results were already known. None of these changes flips L2's verdict, but an examiner who reads the dates can say the test was written after the results.

**3. L2 is judged on a rule you invented, which goes against your own benchmark rule** The thesis rule (25 September) is that every predictor is tested with its published method, with no new detection rules or thresholds. Nanda et al. only describe the weight norm and propose no detection rule. The config says the L2 windows are "own choices … Not from a paper", and they were settled by looking at plots of the old $p = 97$ models. Your note on the extra effort explains honestly how this happened, but it doesn't remove the inconsistency. L2 is the only predictor judged on a rule designed by the researcher, so its FAIL can be read as a failure of that rule rather than of the weight-norm idea.

**4. Both rules fire at times set by the method, not by grokking** The fast and slow averages already cross upwards several times in the first 100 epochs (in seed 0, for example, at about epochs 2, 3 and 42). The crossover rule only reports 113–137 because

it ignores everything before `skip_epochs` = 100. The averages are taken on a log-epoch grid of 40,000 points, and 43.5% of those points lie before epoch 100. So at epoch 113 the "fast" window covers only about 1.5 epochs and the "slow" one about 6, which means the event is basically an epoch-level wiggle. An upward crossing also means the norm is *rising*, which is the memorisation build-up, while the idea being tested is about the norm *falling*. The MA-of-MA crossing lands at 1767–1814 in every seed while grok runs from 6921 to 23747 ($\rho = 0$). Because both events are fixed early, the negative result says little about whether the weight norm carries grok-timing information.

## Moderate worries

**5. One L2 reading in your own analysis passes the frozen test** The analysis script still computes "$\sum w^2$ halfway down from peak" on every run. Its target is $y_{\text{peak}} - 0.5\,(y_{\text{peak}} - y_{\text{final}})$. Under the frozen rule it gets lead_frac $= 0.8$, $\rho = 0.9$ (exactly the cutoff) and mean relative lead $0.162$, so it would PASS if it were scored. You've documented that it is a hindsight reading: it uses the final value at epoch 40,000, the 50% level is arbitrary, and it is not a benchmark result. That's a fair position, and it passes only at the edges. Still, the 26 September table prints it next to the implemented rules, and whether it goes into the thesis is still undecided. An examiner can ask why the one L2 reading that follows grok is set aside, and the answer rests only on the scope rule.

**6. The moving averages use future values** Both implemented rules smooth with `uniform_filter1d`, which is a centred average with the edges padded by copying. Each smoothed value therefore includes epochs that come after it. The halfway reading's 200-epoch smoothing is centred too. The viva notes describe the MA crossover as an "ex-ante trigger", but strictly it isn't one. This doesn't change the verdict, but the wording can be challenged.

**7. Weight decay alone could explain the fall, and there are no comparison runs** Weight decay is $\lambda = 1.0$ in every run and was never changed. Once training loss is near zero, weight decay pulls $\sum w^2$ down whether or not the model is about to generalise, and Nanda et al. themselves say weight decay drives the cleanup. There is no run without grokking and no run with different weight decay, and the 25 September report admits this. Without these, any movement in the norm can't be credited to grokking rather than to the optimiser.

**8. The model isn't exactly Nanda's, so $\sum w^2$ adds up different weights** Appendix A of Nanda et al. lists an attention output matrix $W_O^j$ for each head, and gives the output space as 113 (with '=' treated separately). The shared model has no $W_O$, so attention output goes straight into the residual stream, and its output layer has 114 classes. The config calls the setup "a faithful replication", and the 3 September fidelity check didn't list either difference. Because L2 is a sum over all weights, these differences change exactly the quantity being compared with Nanda's Figure 7.

**9. Five seeds make the test coarse, and a fixed-clock L2 event almost passes** With $n = 5$ the test is coarse. You documented the decision to keep 5 seeds, along with the 21.7% pass rate of a random "always-early" predictor. The L2 data shows the risk. The peak of $\sum w^2$ falls between epochs 1038 and 1061 in every seed, only 23 epochs apart, but it still ranks the seeds with $\rho = 0.9$. It fails only because its mean relative lead is $0.904$, just above $0.9$. Seed 4 also decides several L2 correlations on its own: the final weight norm against grok gives $r = -0.55$ with all seeds and $+0.96$ without seed 4.

**10. Only the total norm is used** The signal adds up every parameter: embeddings, Q/K/V, the MLP with its biases, and the output head. At the end of training, embeddings make up about 23–30% of $\sum w^2$ and the MLP about 38–46%. The per-module curves are saved, and on 3 September they were named as the next step for settling the L2 question, but no rule or analysis uses them. An examiner can ask which weights are supposed to carry the signal, and whether a change in one part is hidden by the rest.

**11. The source behind "L2 as a predictor" isn't in the thesis library** Your notes say Nanda uses the weight norm only descriptively, and that the idea of weight norm as a controlling quantity comes from Liu et al. ("Towards Understanding Grokking" and "Omnigrok"). On 3 September you noted those papers should be added. They still aren't in `04_Literature/Papers` or its README. The thesis therefore cites Nanda for a predictor Nanda never proposed, and leaves out the paper the idea actually comes from.

## Small slips

**12. The viva notes quote numbers from older runs** The ready answers say the crossover fires at "~113–120 (mean ~115)" while grok "ranges up to ~13000", and the 17 September note gives a mean grok of 13,218.8 (range 6,988–24,021). Those numbers come from earlier runs. The final run gives crossover epochs of 113–137 (mean 122.8) and grok epochs of 6,921–23,747 (mean 12,836.8).

**13. The number of possible $\rho$ values is stated wrongly** The criterion-validation record says that with five seeds $\rho$ can only take 11 values, in steps of 0.2. In fact $\rho = 1 - \sum d^2/20$ can take 21 values, in steps of 0.1. That list leaves out $0.9$, which is both the calibrated cutoff and the value the halfway reading reaches.

**14. The "before grok" column uses the wrong margin** The summary table counts a seed as early if its lead is above 0 epochs, not above the protocol's 100-epoch margin. That's why the 26 September table shows the halfway reading as 5/5. Under the protocol it's 4/5, because seed 2 leads by only 13 epochs.

**15. The analysis would crash if the halfway level were never reached** `l2_decline_epoch` takes the first index without checking that one exists, so a run whose norm never falls halfway would stop the whole analysis script. It doesn't happen with the current data.

**16. Code text that no longer matches** The runner's header still says it runs "ONLY L2-Norm and Dropout" and that the later predictors aren't built. That has been known since 25 September. The noise floor is computed and saved with the L2 results but isn't used in any decision. It's measured over epochs before 90, which is mostly interpolated grid points.

## What checks out

The grok epochs (12987, 6921, 9510, 11019, 23747) and both L2 event sets (crossover 113.4 / 113.2 / 120.1 / 130.6 / 136.8, MA-of-MA 1802 / 1767 / 1814 / 1811 / 1801) recomputed from the saved files match `summary.json` and the 25–26 September tables. Under every version of the test (August, 26 September and the frozen rule), both implemented rules FAIL, so the negative verdict doesn't depend on which test is used. `l2_norm.py`, `nanda_unified.yaml`, the shared model and the data code haven't changed since the results were committed (`e1cfa3e`), and the runner's L2 lines are unchanged too, so the saved results match the current code. The setup matches Nanda on $p = 113$, the 30% split, full batch, AdamW with $\lambda = 1.0$ and betas $(0.9, 0.98)$, 40,000 epochs, no LayerNorm, MLP biases and small init. The weight norm rises during memorisation and peaks at about epoch 1040–1060 in every seed, which matches Nanda's description.

## Not verified

- Whether Nanda's $\sum w^2$ includes the embeddings and MLP biases, and why his Figure 7 peak (about 3,600) is lower than the peaks here (about 4,800–5,000).
- What Liu et al. ("Omnigrok") actually claim about weight norm and timing, and whether they give any detection rule, since the paper isn't in the library.